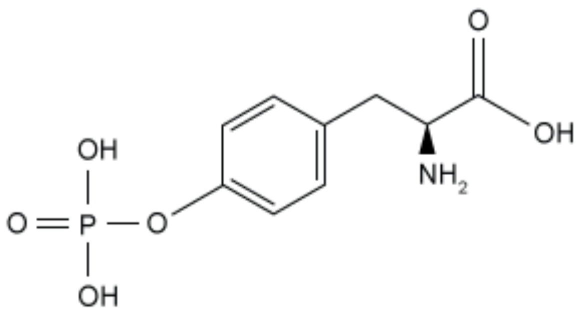
The structure of the molecules in a sample of solid L-phosphotyrosine is shown below.

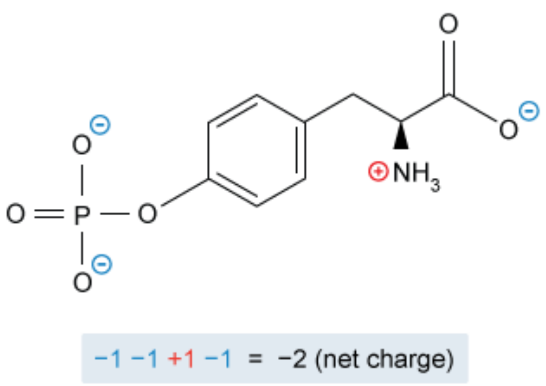


What will be the net charge of the majority of L-phosphotyrosine molecules when placed in an aqueous solution at pH 8.0? (Note: Assume that the pK_a values for the functional groups in the structure are: 2.0 for the $-CO_2H$ group, 9.5 for the $-NH_2$ group, and 2.2 and 7.2 for the $-OPO_3H_2$ group.)

- A. 0 [10%]
- B. −1 [15%]
- C. −2 [52%]
- D. −3 [20%]

Correct
52% answered correctly

Explanation:



The **net charge** of a molecule is the **sum** of all nonzero formal charges from each atom in the structure. Atoms often acquire a **nonzero formal charge** by acid-base interactions in solution. Functional groups that are acidic act as **proton (H^+) donors**, but those that are basic act as proton acceptors. Each proton donated by an atom decreases its formal charge by one unit, but each proton accepted by an atom increases the formal charge by one unit.

The response of an acidic or basic functional group depends on its pK_a relative to the pH of the solution.

- If pK_a,< pH, then the functional group will be deprotonated.
- If pK_a,> pH, then the functional group will be protonated.

At pH 8.0, amino acids have negatively charged carboxylic acid groups and positively charged amino groups. Carboxylic acid groups (weak acids with pK_a,≈ 2) will donate a proton and become deprotonated to form negatively charged carboxylates in a solution with pH = 8 because pK_a,< pH. Conversely, the amino groups (bases with pK_a, ≈ 9.5) will accept protons to form aminium cations in a solution with pH = 8 because pK_a,> pH (ie, the group is less acidic and more basic than the solution).

Groups attached to amino acids can also act as an acid or a base. A phosphate group behaves like an attached molecule of phosphoric acid, and at pH 8.0, it also ionizes to acquire two negative charges because its pK_a values are lower than the pH.

Therefore, in an aqueous solution at pH 8.0, the majority of the L-phosphotyrosine molecules will have two negative charges from the phosphate group, one negative charge from the carboxyl group, and one positive charge from the amino group which sum to a net charge of −2.

(Choice A) Zero net charge would require that no ions formed in the structure (or that the charges formed cancelled out). The acid-base behavior of the phosphate, carboxyl, and amino groups excludes this possibility.

(Choice B) A net charge of −1 would occur if only the amino and the phosphate groups formed ions. However, the acidic nature of carboxylic acids ensures that the carboxyl group will ionize also.

(Choice D) A net charge of −3 would be possible if only the acidic phosphate and carboxyl groups formed ions. The basic nature of amines, however, causes the amino group to be protonated and form a +1 charge.

Educational objective:
Some amino acid functional groups act as proton donors or as proton acceptors. Each proton donated by an atom decreases its formal charge by one unit, but each proton accepted by an atom increases the formal charge by one unit. The net charge is the sum of all nonzero formal charges in a molecule.

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